

YUCT and Dissipative Structures

A Coordination-Based Interpretation of Non-Equilibrium Instabilities

Connecting the Universal Error Law to the Onset of Convection

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Abstract

The Yakushev Unified Coordination Theory (YUCT) describes ordered complexity through a single parameter, the coordination efficiency K_{eff} , and a universal error-scaling law $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$ ($\beta = 2/3$, $\kappa_c = 1/3$). This Appendix explores the consequences of this law for the stability of non-equilibrium stationary states, in particular for the onset of Bénard convection. We propose a definition of K_{eff} for a simple fluid in terms of its heat conductivity and shear viscosity—quantities that are routinely measurable. Using this definition, we express the entropy production rate σ as a function of K_{eff} and show that the linear stability threshold corresponds to a universal critical value of the coordination efficiency, $K_{\text{eff}}^{\text{crit}} = (S_{\text{odd}}/S_{\text{even}})^{1/\beta} \approx 1.837$. This value follows directly from the algebraic loop of YUCT ($S_{\text{odd}} = 6/5$, $S_{\text{even}} = 4/5$, $\beta = 2/3$) and contains no free parameters. The prediction is that for any Newtonian fluid, convection sets in when its effective coordination efficiency, as defined here, drops below ≈ 1.837 . We derive a quantitative, falsifiable relation between the critical Rayleigh number and the fluid's Prandtl number, which can be tested with existing laboratory data. The derivation makes no use of the KPZ relation or fractal dimensions; it relies solely on the algebraic structure of YUCT. Limitations of the present phenomenological treatment are explicitly discussed.

Keywords: YUCT, Bénard convection, coordination efficiency, entropy production, Prandtl number, algebraic loop.

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1 Introduction

The Yakushev Unified Coordination Theory (YUCT) is built upon the premise that coordination—the ability of a system’s components to align their states using pre-existing shared dictionaries—is the fundamental process from which physical laws emerge [1]. Its predictive power rests on a single scaling law for the relative error of coordination:

$$\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}, \quad \beta = \frac{2}{3}, \quad \kappa_c = \frac{1}{3}, \quad (1)$$

with α a system-specific constant of order unity. The values of β and κ_c are not free parameters; they follow from the algebraic self-consistency of hierarchical coordination networks, as detailed in Appendices [2, 3, 4, 5]. The resulting algebraic loop yields two exact rational constants,

$$S_{\text{odd}} = \frac{6}{5} = 1.2, \quad S_{\text{even}} = \frac{4}{5} = 0.8, \quad (2)$$

which satisfy $S_{\text{odd}} + S_{\text{even}} = 2$ and $S_{\text{odd}}/S_{\text{even}} = 3/2 = 1/\beta$. These numbers are the backbone of all subsequent derivations.

In a different line of research, the theory of dissipative structures formulated by Ilya Prigogine describes how open systems driven far from thermodynamic equilibrium can spontaneously evolve towards states of higher complexity [6, 7]. The archetypal example is Bénard convection: a horizontal fluid layer heated from below develops a regular pattern of convective rolls when the temperature gradient exceeds a critical value [8]. Prigogine’s stability criterion states that a stationary state becomes unstable when the excess entropy production $\delta_X P$ becomes positive.

The purpose of this Appendix is to show that Prigogine’s criterion can be reformulated in the language of YUCT, and that the algebraic loop of YUCT makes a quantitative, testable prediction for the onset of convection. The derivation is phenomenological, but it contains no arbitrary parameters beyond those already present in the YUCT framework. We also explicitly acknowledge the limitations of the present approach and outline the steps required for a full microscopic theory.

2 Coordination Efficiency of a Simple Fluid

To apply YUCT to a physical system, one must identify the relevant “dictionary” and “index”, and from them compute K_{eff} . For a fluid in a temperature gradient, the transport of heat can be viewed as a series of local coordination events: molecules at a higher temperature transfer energy to those at a lower temperature via collisions. The efficiency of this process is limited by two factors: the intrinsic ability of the medium to conduct heat (its thermal conductivity λ) and the internal friction that dissipates energy (its shear viscosity η). A fluid with high λ and low η is a “good coordinator” because energy is moved quickly with little loss.

We therefore propose the following operational definition:

$$K_{\text{eff}} \equiv \frac{\lambda}{\lambda_0} \left(\frac{\eta_0}{\eta} \right)^\gamma, \quad (3)$$

where λ_0 and η_0 are reference values (e.g. for water at 20°C) and the exponent γ is chosen such that K_{eff} becomes a dimensionless, scale-independent quantity. Since the ratio λ/η has the

dimension of $(\text{length})^2/(\text{time} \cdot \text{temperature})$, a natural choice is $\gamma = 1$, giving

$$K_{\text{eff}} = \frac{\lambda/\lambda_0}{\eta/\eta_0}. \quad (4)$$

This definition is analogous to the “coordination number” introduced for chemical elements in Appendix C, and it makes K_{eff} directly measurable.

3 Entropy Production and the Universal Error Law

For a fluid at rest in a temperature gradient ∇T , the sole thermodynamic flux is the heat flux $J = -\lambda \nabla T$, and the conjugate force is $X = \nabla(1/T)$. The entropy production rate per unit volume is

$$\sigma = J \cdot \nabla \left(\frac{1}{T} \right) \approx \frac{\lambda |\nabla T|^2}{T^2}. \quad (5)$$

In the YUCT picture, the heat flux is the result of many microscopic coordination acts. A fraction ε of these acts fail to transfer energy coherently, so the effective flux is reduced:

$$J_{\text{eff}} = J (1 - \varepsilon). \quad (6)$$

The “lost” flux contributes to entropy production but not to the coherent removal of the gradient. Using the universal error law (1), we can write

$$\sigma = \frac{\lambda |\nabla T|^2}{T^2} \frac{1}{1 - \kappa_c \alpha K_{\text{eff}}^{-\beta}}. \quad (7)$$

For $K_{\text{eff}} \gg 1$ the correction is negligible, and the system obeys the usual linear Fourier law. As K_{eff} decreases (the fluid becomes more viscous relative to its conductivity), the error grows, and the entropy production increases for the same imposed gradient.

4 Stability Threshold and the Critical Coordination Efficiency

Prigogine’s stability criterion states that a stationary state becomes unstable when a fluctuation causes the excess entropy production to become positive [7]. In the YUCT formulation, a fluctuation can be thought of as a local change in the coordination state of the fluid—a temporary deviation of K_{eff} from its stationary value. Using (7), a small variation δK_{eff} leads to a change in the entropy production:

$$\delta_X P \propto -\delta K_{\text{eff}}. \quad (8)$$

Thus, a fluctuation that *decreases* K_{eff} makes $\delta_X P > 0$ and tends to be amplified. This amplification accelerates the degradation of coordination, driving the system towards a new state.

The stationary conductive state therefore becomes unstable when K_{eff} falls below a critical value. Because the YUCT algebraic loop fixes all dimensionless ratios, this critical value must be expressible in terms of the fundamental constants:

$$K_{\text{eff}}^{\text{crit}} = \left(\frac{S_{\text{odd}}}{S_{\text{even}}} \right)^{1/\beta} = \left(\frac{3}{2} \right)^{3/2} \approx 1.837. \quad (9)$$

The exponent $1/\beta$ follows from the requirement that the error parameter ε at the threshold be of order unity (the system can no longer distinguish useful coordination from noise). The ratio $S_{\text{odd}}/S_{\text{even}} = 3/2$ is the universal order-disorder ratio of YUCT. Equation (9) contains no free parameters.

5 Prediction for the Critical Rayleigh Number

The Rayleigh number for a fluid layer of depth d , density ρ , specific heat c_p , thermal expansion coefficient α_T , and imposed temperature difference ΔT is

$$\text{Ra} = \frac{\rho^2 c_p \alpha_T g \Delta T d^3}{\lambda \eta}. \quad (10)$$

In terms of our coordination efficiency, (4), we can write

$$\text{Ra} = \frac{A}{K_{\text{eff}}}, \quad (11)$$

where A is a constant that depends only on the geometry and the boundary conditions, not on the fluid properties. With this identification, the threshold condition $K_{\text{eff}} = K_{\text{eff}}^{\text{crit}}$ translates into a critical Rayleigh number:

$$\text{Ra}_c = \frac{A}{K_{\text{eff}}^{\text{crit}}}. \quad (12)$$

If the constant A can be determined from one reference fluid, say water, for which the classic critical Rayleigh number is $\text{Ra}_{\text{water}} \approx 1708$ (for rigid boundaries) and our definition gives some $K_{\text{eff}}^{\text{water}}$, then the critical Rayleigh number for any other fluid can be predicted:

$$\boxed{\text{Ra}_c(\text{fluid}) = \text{Ra}_c(\text{water}) \frac{K_{\text{eff}}^{\text{water}}}{K_{\text{eff}}(\text{fluid})}}. \quad (13)$$

Using (4), this becomes a relation between the critical Rayleigh number and the fluid's thermal conductivity and viscosity:

$$\text{Ra}_c \propto \frac{\eta/\eta_0}{\lambda/\lambda_0}. \quad (14)$$

Thus, a fluid with higher viscosity (poorer coordination) should have a *larger* critical Rayleigh number—it is harder to make it convect. This prediction is qualitatively consistent with classical results: oils, which are viscous, have much higher critical Rayleigh numbers than water. The quantitative test requires measuring λ and η for the fluid, computing K_{eff} from (4), and comparing the predicted Ra_c with the experimentally observed onset of convection. No adjustable parameters remain after the single calibration for the reference fluid.

6 Testable Consequences and Experimental Protocol

6.1 Immediate test with existing data

The literature contains extensive tables of critical Rayleigh numbers for various fluids (see, e.g., the reviews by [8, 9]). Using measured values of λ and η at the relevant temperatures, one can compute K_{eff} for each fluid and verify whether the ratio $\text{Ra}_c/\text{Ra}_c^{\text{water}}$ equals $K_{\text{eff}}^{\text{water}}/K_{\text{eff}}$. A positive correlation with $R^2 > 0.9$ would provide strong evidence for the YUCT mechanism.

6.2 Dedicated experiment

A controlled experiment could be performed with two fluids that have widely different coordination efficiencies, for example water and a silicone oil. For each fluid, the critical temperature difference ΔT_c is measured with high precision. The YUCT prediction is that the ratio of the critical temperature differences (at fixed geometry) should equal the inverse ratio of the coordination efficiencies as defined by (4). This is a parameter-free, falsifiable prediction.

7 Limitations and Open Questions

The present analysis is phenomenological. A full microscopic theory would need to derive the connection between the coordination error ε and the thermodynamic fluxes from first principles, without postulating (6). Furthermore, the definition of K_{eff} in (4) is a motivated ansatz; other definitions may be possible and should be tested against the data. The analysis also ignores the role of chemical reactions, which are known to strongly modify the onset of convection. These are subjects for future research.

Despite these limitations, the algebraic derivation of a critical coordination efficiency $K_{\text{eff}}^{\text{crit}} \approx 1.837$ from the YUCT loop is exact within the framework. Whether nature actually uses this value can be decided by experiment.

8 Conclusions

We have shown that the YUCT framework offers a quantitative, testable interpretation of the onset of Bénard convection. The critical Rayleigh number is expressed in terms of the fluid's coordination efficiency, which itself is linked to measurable transport coefficients. The predicted scaling relation $\text{Ra}_c \propto \eta/\lambda$ can be verified with existing laboratory data. This work demonstrates that YUCT is not merely a philosophical reinterpretation of thermodynamics but a framework capable of generating falsifiable numerical predictions.

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